

DAY 3 INTRODUCTION TO IP ADDRESS IPv4 AND IPv6

List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

ANS:

Here is a comparison table outlining the key differences between IPv4 and IPv6 addressing:

IPv4 Addressing	IPv6 Addressing
1. Address Length: Uses 32-bit addresses, which limits the total number of unique addresses.	1. Address Length: Uses 128-bit addresses, providing an astronomically large pool of unique addresses.
2. Address Format: Represented as four decimal numbers separated by dots (e.g., 192.0.2.1).	2. Address Format: Represented as eight groups of hexadecimal values separated by colons (e.g., 2001:db8::1).
3. Address Exhaustion: Suffers from address depletion, heavily relying on Network Address Translation (NAT) to share public IPs.	3. Address Exhaustion: Practically inexhaustible, eliminating the structural need for NAT.
4. Configuration: Requires manual configuration, DHCP, or DHCPv4 for a device to get an IP address.	4. Configuration: Supports Stateless Address Autoconfiguration (SLAAC) and DHCPv6 natively for plug-and-play setup.

Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

ANS:

Here is the static IPv4 configuration for your 4 home devices using the **192.168.10.0/24** network (where **192.168.10.1** is typically reserved as the default gateway/router):

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

ANS:

Python

```
import ipaddress
```

```
network = ipaddress.ip_network('192.168.1.0/24')
```

```
subnets = list(network.subnets(new_prefix=27))
```

```
print(f"Total subnets: {len(subnets)}")
```

```
for i in range(min(3, len(subnets))):
```

```
    sub = subnets[i]
```

```
    net_addr = sub.network_address
```

```
    broadcast = sub.broadcast_address
```

```
    hosts = list(sub.hosts())
```

```
    print(f"Subnet {i+1}:")
```

```
    print(f" Network Address: {net_addr}")
```

```
    print(f" Broadcast Address: {broadcast}")
```

```
    print(f" Usable Host Range: {hosts[0]} - {hosts[-1]}")
```

```
    print(f" Total Usable Hosts: {len(hosts)}")
```

Code output

Total subnets: 8

Subnet 1:

Network Address: 192.168.1.0

Broadcast Address: 192.168.1.31

Usable Host Range: 192.168.1.1 - 192.168.1.30

Total Usable Hosts: 30

Subnet 2:

Network Address: 192.168.1.32

Broadcast Address: 192.168.1.63

Usable Host Range: 192.168.1.33 - 192.168.1.62

Total Usable Hosts: 30

Subnet 3:

Network Address: 192.168.1.64

Broadcast Address: 192.168.1.95

Usable Host Range: 192.168.1.65 - 192.168.1.94

Total Usable Hosts: 30

Here are the subnetting calculations for the 192.168.1.0 network using a /27 CIDR subnet mask (Subnet Mask: 255.255.255.224):

Summary Statistics

- **Number of Valid Subnets: 8 subnets** (2^3 bits borrowed from the host portion)
-
- **Hosts per Subnet: 30 usable hosts** ($2^5 - 2$ available host bits)

First Three Valid Subnet Ranges

Subnet #	Network Address	Usable Host Range	Broadcast Address
Subnet 1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
Subnet 2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
Subnet 3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.

ANS:

Python

```
import ipaddress
```

```
ip = ipaddress.ip_interface('10.0.5.17/255.255.255.0')
network = ip.network
print(f"Network Address: {network.network_address}")
print(f"Broadcast Address: {network.broadcast_address}")
print(f"Usable Host Range: {network.network_address + 1} - {network.broadcast_address - 1}")
```

Code output

```
Network Address: 10.0.5.0
Broadcast Address: 10.0.5.255
Usable Host Range: 10.0.5.1 - 10.0.5.254
```

Here are the calculations for the IP address 10.0.5.17 with the subnet mask 255.255.255.0 (which corresponds to /24 CIDR notation):

- **Network Address:** 10.0.5.0
-
- **Broadcast Address:** 10.0.5.255
-
- **Range of Valid Host Addresses:** 10.0.5.1 to 10.0.5.254

Use an online subnet calculator tool (such as <https://www.subnet-calculator.com/>) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

ANS:



Subnet Calculator

Network Class

A ☒ B ☐ C ☐

First Octet Range

1 - 126

IP Address

10.0.0.1

Hex IP Address

0A.00.00.01

Subnet Mask

255.0.0.0



Wildcard Mask

0.255.255.255

Subnet Bits

0



Mask Bits

8



Maximum Subnets

1



Hosts per Subnet

16777214



Host Address Range

10.0.0.1 - 10.255.255.254

Subnet ID

10.0.0.0

Broadcast Address

10.255.255.255

Subnet Bitmap

0nnnnnnnn.hhhhhhhh.hhhhhhhh.hhhhhhhh